6.2 popquiz complex circuit - test review

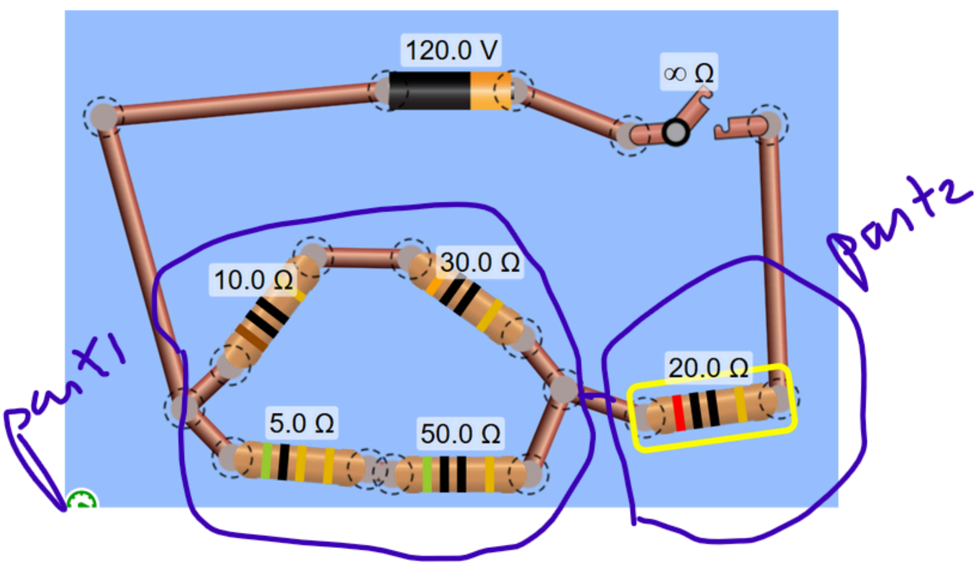
This is a preview of the published version of the quiz

Started: Mar 22 at 7:05pm

Quiz Instructions

You can check your answers with the simulation. But no simulation for the test. So try without first.

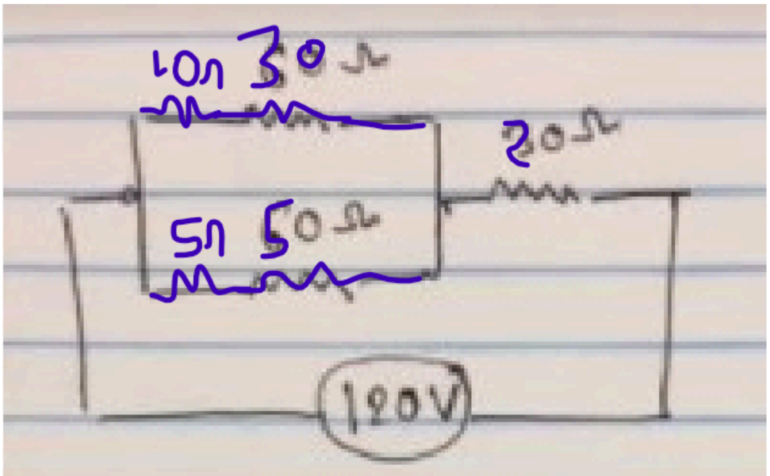
Question 1 20 pts



Consider this circuit. What is the equivalent resistance ? you have 2 parts in series with each other.

the 2 branches of part 1 are in parallel.

Here to help you. The schematic:



- about 43 ohms
- about 23 ohms
- about 115 ohms
- about 10.7 ohms

Question 2 20 pts

Building on previous question. What is the main current? The current drained from the battery.

you can check with the simulation but first try without. There is no simulation for the test.

https://phet.colorado.edu/sims/html/circuit-construction-kit-dc/latest/circuit-construction-kit-dc_all.html (https://phet.colorado.edu/sims/html/circuit-construction-kit-dc/latest/circuit-construction-kit-dc_all.html)

- about 1 A
- about 11.2 A
- about 5.2 A
- about 2.8 A

Question 3 20 pts

Build on previous question. What is the voltage across the 20 ohms?

easy. Use Ohms law. V = R I and I is the main current.

- about 57 V
- about 20V
- about 104V
- about 224V

Question 4 20 pts

So you have Vpart1 + V part2 = 120V and you have Vpart2.

What is the voltage across V part1 ?

- about 24 V
- about 63V
- about 100V
- about 16V

Question 5 20 pts

Building on question. The 2branches of part 1 are in parallel so they have the same voltage. You computed the voltage across part1 in the previous question. Vpart1.

What is the current I1 through the branch 10ohms + 30ohms (in series with each other)

Use ohms law. I1 = Vpart1/ (10+30)

- about 0.4A
- about 2.5A
- about 1.6 A
- about 0.6A

Question 6 5 pts

building on the previous question. Now you have the current through the top branch 10ohms + 30ohms.

What is the voltage across the 30ohms? (hint so use Ohms law V30 = 30 x I1. I1 is the current through the branch computed previously

- about 18V
- about 48V
- about 75V
- about 12V

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